

Chapter 1. Finite Groups

Part II - Characters, Tensor Products and Young Tableaux

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Characters

Character

The **characters** $\chi_D(g)$ of a group rep D are defined as

$$\chi_D(g) \equiv \text{tr } D(g) = \sum_i [D(g)]_{ii}. \quad (1)$$

By cyclic property of traces $\text{tr}(AB \cdots) = \text{tr}(B \cdots A)$, characters are *unchanged* under similarity transformations :

$$\text{tr}(S^{-1}DS) = \text{tr}(SS^{-1}D) = \text{tr}(D)$$

Equivalent reps have the same characters.

e.g. Regular rep's character is $\chi_R(e) = |G|$, $\chi_R(g) = 0$ if $g \neq e$

Orthogonality of Characters

From Schur orthogonality relation

$$\frac{1}{|G|} \sum_{g \in G} [D_a(g)]_{jk}^* [D_b(g)]_{lm} = \frac{\delta_{ab} \delta_{jl} \delta_{km}}{n_a}$$

Let $j = k$, $l = m$ and summing over :

$$\frac{1}{|G|} \sum_{g \in G} \sum_j \sum_l [D_a(g)]_{jj}^* [D_b(g)]_{ll} = \frac{1}{|G|} \sum_{g \in G} \chi_{D_a}(g)^* \chi_{D_b}(g) \quad (2)$$

$$= \delta_{ab} \quad (3)$$

Therefore, characters of different irreps are **orthogonal**

Conjugacy Classes

Conjugacy Classes

A **conjugacy class** of $x \in G$, denoted $\text{Cl}(x)$, is

$$\text{Cl}(x) = \{gxg^{-1} | g \in G\}. \quad (4)$$

Conjugacy classes form a partition of group G .

- Conjugacy class of x can be thought as an 'orbit' of x under conjugation action of G on itself :

$$\text{Cl}(x) = \{x, g_1 x g_1^{-1}, g_2 x g_2^{-1}, \dots\}, g_i \in G$$

e.g. Conjugacy classes of $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ are $\{0\}, \{1\}, \dots, \{n-1\}$, since $\mathbb{Z}_n$ is abelian : $g + x + (-g) = x \quad \forall g, x \in \mathbb{Z}_n$

Characters on Conjugacy Classes

The characters are **constant** on conjugacy classes because if $g_2 \in \text{Cl}(g_1)$,

$$\begin{aligned}\text{tr } D(g_2) &= \text{tr } D(g^{-1}g_1g) = \text{tr } D(g)^{-1}D(g_1)D(g) \\ &= \text{tr } D(g_1) \quad \text{For some } g \in G.\end{aligned}$$

*Question : We saw that any function of G , $F(g)$ can be expanded in $[D_a(g)]_{jk}$. Can any function of G , which is **constant** on the conjugacy classes, be expanded in $\chi_D(g)$?*

Completeness of Characters

Suppose $F(g)$ is a function of G such that

$$g_2 \in \text{Cl}(g_1) \implies F(g_1) = F(g_2)$$

Again using Schur orthogonality, $F(g_1)$ can be expanded in terms of irrep $D_a(g)$'s matrix elements

$$F(g_1) = \sum_{a,j,k} c_{jk}^a [D_a(g_1)]_{jk}$$

Using the fact that F is constant on conjugacy classes

$$F(g_1) = \frac{1}{|G|} \sum_{g \in G} F(g^{-1} g_1 g) = \frac{1}{|G|} \sum_{g \in G} \sum_{a,j,k} c_{jk}^a [D_a(g^{-1} g_1 g)]_{jk}$$

Completeness of Characters

Using Schur orthogonality,

$$\begin{aligned} \frac{1}{|G|} \sum_{g \in G} \sum_{a,j,k} \sum_{l,m} c_{jk}^a [D_a(g^{-1})]_{jl} [D_a(g_1)]_{lm} [D_a(g)]_{mk} \\ = \sum_{a,j,k} \sum_{l,m} c_{jk}^a [D_a(g_1)]_{lm} \frac{1}{n_a} \delta_{jk} \delta_{lm} \\ \therefore F(g_1) = \sum_{a,j} \frac{1}{n_a} c_{jj}^a \chi_{D_a}(g_1) \end{aligned} \quad (5)$$

Therefore the characters $\chi_{D_a}(g)$ form a **complete, orthonormal basis set** for the functions that are constant on conjugacy classes

Cor. (# of inequivalent irreps of G) = (# of conjugacy classes of G)

Another Orthogonality Relation

Character orthogonality relations (sum over $g \in G$)

$$\sum_{g \in G} \chi_{D_a}(g)^* \chi_{D_b}(g) = |G| \delta_{ab}$$

Labeling conjugacy classes as α , with k_α elements and letting $a = b$,

$$\sum_{\alpha} k_{\alpha} \chi_{D_a}(g)^* \chi_{D_a}(g) = |G|$$

Define matrix elements

$$V_{\alpha a} \equiv \sqrt{\frac{k_{\alpha}}{|G|}} \chi_{D_a}(g_{\alpha}), \quad g_{\alpha} = \text{Conjugacy class}$$

Character Orthogonality Relation over Irreps

Then our character orthogonality relation is just

$$V^\dagger V = 1$$

Since V is unitary, we also have $VV^\dagger = 1$

$$\frac{1}{|G|} \sum_a \chi_{D_a}(g_\alpha)^* \chi_{D_a}(g_\beta) = \frac{\delta_{\alpha\beta}}{k_\alpha}$$

This is a character orthogonality relation over irrep labels a

Multiplicity

Suppose D is an completely reducible into irreps D_a , with **multiplicity** m_a :

$$D = \bigoplus_a \underbrace{(D_a \oplus D_a \oplus \cdots D_a)}_{m_a \text{ times}}$$

The character orthogonality relation provides a measure of *how many times a given irrep appears* in a representation :

$$\frac{1}{|G|} \sum_{g \in G} \chi_{D_a}(g)^* \chi_D(g) = m_a \quad (6)$$

e.g. For an irrep D_a of regular rep, multiplicity is n_a

Summary of Orthogonality Relations

① Schur Orthogonality

$$\frac{1}{|G|} \sum_{g \in G} [D_a(g)]_{jk}^* [D_b(g)]_{lm} = \frac{\delta_{ab} \delta_{jl} \delta_{km}}{n_a}$$

② Character Orthogonality (over G)

$$\sum_{g \in G} \chi_{D_a}(g)^* \chi_{D_b}(g) = |G| \delta_{ab}$$

③ Character Orthogonality (over irreps)

$$\frac{1}{|G|} \sum_a \chi_{D_a}(g_\alpha)^* \chi_{D_a}(g_\beta) = \frac{\delta_{\alpha\beta}}{k_\alpha}$$

Example : S_3

The **permutation group** S_3 consists of all possible permutations of the set $\{1, 2, 3\}$

We often write S_3 as

$$S_3 = \{e, (123), (321), (12), (23), (31)\}$$

where (123) denotes a permutation $1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 1$, and (12) denotes an exchange $1 \leftrightarrow 2$, fixing 3

Mission objectives :

- 1 Find conjugacy classes of S_3 .
- 2 Determine characters of S_3 .
- 3 Reduce a 3-dimensional rep of S_3 into irreps.

Mission 1. Conjugacy Classes of S_3

How many irreps?

→ Let's determine number of **conjugacy classes**!

① $\{e\}$ itself is a conjugacy class

② Observing that

$$(12)^{-1}(123)(12) = (321) \quad (123) \sim (321)$$

We can easily conclude another conjugacy class $\{(123), (321)\}$

③ One can find $\{(12), (23), (31)\}$ as the remaining conjugacy class

Note : There is a systematic way to determine conjugacy classes, known as the Young tableaux.

Conclusion : there are 3 irreps of S_3 !

Mission 2. Characters of S_3

- 1 The trivial rep
Character 1, since all $D(g) = 1$

- 2 The nontrivial one ('sign' rep)
Let's define one-dim representation of S_3 as follows:

$$D(g) = \begin{cases} 1 & g \in \{e, (123), (321)\} \\ -1 & g \in \{(12), (23), (31)\} \end{cases}$$

Then this is an irrep. The characters are, of course,

conj.class	e	$(123), (321)$	$(12), (23), (31)$
χ	1	1	-1

Mission 2. Characters of S_3

So far we have determined the character table as follows:

irrep/conj	e	$(123), (321)$	$(12), (23), (31)$
0	1	1	1
1	1	1	-1
2	2	a	b

Now we use orthogonality of characters, eqn [3]

$$2 + 2\mathbf{a} + 3\mathbf{b} = 0$$

$$2 + 2\mathbf{a} - 3\mathbf{b} = 0$$

This gives $a = -1$ and $b = 0$

Mission 2. Characters of S_3

irrep/conj	e	$(123), (321)$	$(12), (23), (31)$
0	1	1	1
1	1	1	-1
2	2	-1	0

Table 1: Character table of S_3 .

Now we can use this table to decompose a rep D into irreps!

$$\frac{n_a}{|G|} \sum_{g \in G} \chi_{D_a}(g)^* \chi_D(g) = \delta_{ab}$$

The blue part corresponds to a **projection operator** P_a onto irrep D_a

Mission 3. Decomposing $D(S^3)$

Let's construct one rep of S_3 , D_3 as follows :

- 1 Let $\{1, 2, 3\}$ be $|1\rangle, |2\rangle, |3\rangle$.
- 2 Define the 'action' of $\sigma \in S_3$ as 'basis change'

$$(12) \in S_3 \quad \leftrightarrow \quad |1\rangle \leftrightarrow |2\rangle$$

Then we can easily identify, for example,

$$D(12) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad D(12) |1\rangle = |2\rangle$$

Now let's decompose D_3 into irreps!

Mission 3. Decomposing $D(S^3)$

$$D(e) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D((12)) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$D((13)) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad D((23)) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$D((123)) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad D((132)) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Mission 3. Decomposing $D(S^3)$

$$P_0 = \frac{1}{6} \left[\sum_{g \in G} D_3(g) \right] = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

$$P_1 = 0$$

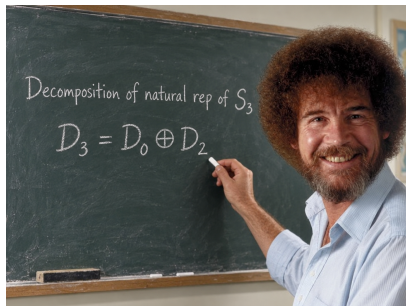
$$P_2 = \frac{2}{6} \left[2D_3(e) + (-1)(D_3(123) + D_3(321)) \right] = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

Then P_0 projects onto the subspace $\text{span} \{ |1\rangle + |2\rangle + |3\rangle \}$, while P_2 projects onto $\text{span} \{ |1\rangle - |2\rangle, |3\rangle - |2\rangle \}$

Mission 3. Decomposing $D(S^3)$

Therefore we've reduced our D_3 into irreps!

$$D_3 = D_0 \oplus D_2 \quad (7)$$



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Eigenstates

What is the role of our 'commuting observable' H , especially when H is hermitian?

$$[H, D(g)] = 0 \quad \forall g \in G$$

By **spectral theorem**, we can 'divide up' the Hilbert space into subspaces V_h , with different eigenvalues h of H :

$$V = \bigoplus_h V_h, \quad (8)$$

Let $|h\rangle$ is such that $H |h\rangle = h |h\rangle$

$$HD(g) |h\rangle = D(g)H |h\rangle = hD(g) |h\rangle, \quad \forall g \in G$$

Therefore $D(g) |h\rangle \in V_h \quad \forall g \in G$, and V_h are **invariant subspaces**

Eigenstates

So an invariant subspace V_h can be decomposed into irreps :

$$V_h = \bigoplus_{\mu} V_{h,\mu}$$

Question : What is the 'form' of H ?

If we assume that irreps appear only once, by **Schur's lemma**

$$HD_a(g) = D_a(g)H \rightarrow H \propto h_a I$$

So If

$$D(g) = \bigoplus_i D_i(g) \quad , \quad [H, D(g)] = 0$$

Eigenstates

$$H = \begin{pmatrix} h_1 \mathbf{I}_{n_1} & 0 & 0 & 0 & \cdots \\ 0 & h_2 \mathbf{I}_{n_2} & 0 & 0 & \cdots \\ 0 & 0 & h_3 \mathbf{I}_{n_3} & 0 & \cdots \\ 0 & 0 & 0 & h_4 \mathbf{I}_{n_4} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

n_a -fold degeneracy - H can't distinguish states within an irrep!

e.g. Hydrogen atom.

Let us denote a state of a Hydrogen atom as $|n, l, m\rangle$ (ignoring spin)

l labels an irrep of $SO(3)$, so states with fixed l , labeled by $-l, \dots, l$ have the same energy eigenvalue - **a $(2l + 1)$ -fold degeneracy!**

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Tensor Products

Can we build a bigger representation from the old ones?

Tensor Products

Let D_1 be an m -dim rep of G , acting on V and D_2 be an n -dim rep of G , acting on W . Then the **tensor product** of D_1 and D_2 , denoted $D_1 \otimes D_2$, is a rep acting on $V \otimes W^a$ such that

$$[D_1 \otimes D_2](g) = D_1(g) \otimes D_2(g).$$

^aFor much details, see [45]

If we take two bases $\{|j\rangle\}$ and $\{|x\rangle\}$, we may form a new basis

$$|j, x\rangle = |j\rangle \otimes |x\rangle$$

, spanning an mn -dimensional vector space

Tensor Products

- Bilinearity :

$$[D_1(g) \otimes D_2(g)][|j\rangle \otimes |x\rangle] = D_1(g) |j\rangle \otimes D_2(g) |x\rangle$$

Using bilinearity, we can compute matrix element

$$\begin{aligned}\langle j, x | D_1(g) \otimes D_2(g) | k, y \rangle &= \left[\langle j | \otimes \langle x | \right] \left[D_1(g) \otimes D_2(g) \right] \left[| k \rangle \otimes | y \rangle \right] \\ &= \left[\langle j | \otimes \langle x | \right] \left[D_1(g) | k \rangle \otimes D_2(g) | y \rangle \right] \\ &= \langle j | D_1(g) | k \rangle \langle x | D_2(g) | y \rangle\end{aligned}$$

Tensor Products

Schematically, if

$$D_A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, D_B = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$$

Then the tensor product $D_A \otimes D_B$ looks like

$$D_A \otimes D_B = \begin{pmatrix} aD_B & bD_B \\ cD_B & dD_B \end{pmatrix} = \begin{pmatrix} ap & aq & bp & bq \\ ar & as & br & bs \\ cp & cq & dp & dq \\ cr & cs & dr & ds \end{pmatrix}$$

Even if D_A and D_B are irreps, $D_A \otimes D_B$ can be *reducible*!

Decomposing $D_A \otimes D_B$ into irreps is known as the **Clebsch-Gordan decomposition**

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Permutation Group

Permutation Group

Let S be a finite set with n elements. A set of all bijections $f : S \rightarrow S$ forms a group under function composition $\circ$. This group is called the **permutation group** and denoted S_n .^a

^aA permutation group can be defined on an infinite set S identically.

For a set $S = \{s_1, s_2, \dots, s_n\}$, we usually write $\sigma \in S_n$ as

$$\sigma \equiv \begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \cdots & \sigma(n) \end{pmatrix}$$

e.g. For $\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$, $\tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$ in S_3 , $\tau\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$ and $\sigma\tau = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$

Permutation Group

Cycles

An n -**cycle** $(i_1 i_2 \cdots i_n)$ is an element of S_n such that $i_1 \mapsto i_2, i_2 \mapsto i_3, \dots, i_n \mapsto i_1$. A 1-cycle is an identity map. A 2-cycle is called a **transposition**.

e.g. $\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = (123), \tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = (13)(2).$

Theorem

Any permutation can be decomposed into products^a of transpositions.

^aNot necessarily unique.

We can find a simple representation of S_n , by taking set S 's elements to form a basis : $|1\rangle, |2\rangle, \dots, |n\rangle$

Permutation Group

Then if a permutation σ takes $i \rightarrow j$,

$$D(\sigma) |i\rangle = |\sigma(i)\rangle = |j\rangle$$

$$\langle l | D(\sigma) | i \rangle = \delta_{lj}$$

e.g. $\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$'s matrix representation is

$$\begin{pmatrix} \langle 1 | \sigma(1) \rangle & \langle 1 | \sigma(2) \rangle & \langle 1 | \sigma(3) \rangle \\ \langle 2 | \sigma(1) \rangle & \langle 2 | \sigma(2) \rangle & \langle 2 | \sigma(3) \rangle \\ \langle 3 | \sigma(1) \rangle & \langle 3 | \sigma(2) \rangle & \langle 3 | \sigma(3) \rangle \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Note that each matrix has a single 1 in each row and column

Conjugacy Classes of the Permutation Group

Generally, we can label 'cycle structure' of a permutation σ :

- If σ has k_j j -cycles, where

$$\sum_{j=1}^n j k_j = n,$$

Then σ has a cycle structure $(k_1 k_2 \cdots)$

Question : Can a conjugation operation $\sigma \rightarrow g\sigma g^{-1}$ change cycle structure of σ ?

Let $\sigma = (1)(\mathbf{23})(4)$, and let's compute conjugate action by $(12)(3)(4)$

$$(12)(3)(4)(1)(\mathbf{23})(4)(12)(3)(4)$$

Conjugacy Classes of the Permutation Group

$$\begin{aligned} & [(12)(3)(4)(\mathbf{1})(\mathbf{23})(4)(12)(3)(4)](1) \\ &= [(12)(3)(4)(\mathbf{1})(\mathbf{23})(4)](2) \\ &= [(12)(3)(4)](3) \\ &= 3 \end{aligned}$$

Therefore one can find $(12)(3)(4)(\mathbf{1})(\mathbf{23})(4)(12)(3)(4) = (2)(13)(4)$
→ cycle structure is unchanged!

In terms of basis $|1\rangle, |2\rangle, \dots$, a conjugation $\sigma \rightarrow g\sigma g^{-1}$ is 're-label' of the basis

$$(|1\rangle, |2\rangle, \dots) \rightarrow (|g(1)\rangle, |g(2)\rangle, \dots)$$

Conjugacy Classes of the Permutation Group

Then it is almost clear that a cycle structure is conserved under conjugation:

$$(123)(45) \rightarrow (g(1)g(2)g(3))(g(4)g(5))$$

Also, conjugacy class will be

$$(\text{Conjugacy class of } (123)(45)) = \bigcup_{\text{All permutations}} (g(1)g(2)g(3))(g(4)g(5))$$

The **conjugacy classes** must consist of **all possible permutations** with a particular cycle structure

The number of group elements in each conjugacy class labeled by $(k_1 k_2 \cdots)$:

$$\# = \frac{n!}{\prod_j j^{k_j} k_j!} \tag{9}$$

Young Tableaux

We now represent conjugacy classes, using a diagram called **Young tableau**

- ① A j -cycle is a column of boxes of length j
- ② A diagram is arranged in order of decreasing j , as going to right

e.g.

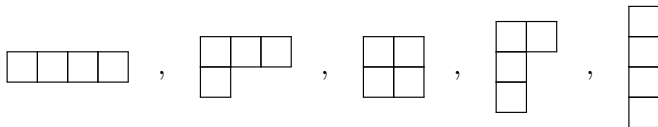
$$(123)(45) \in \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \\ \hline \end{array}, \quad (1234)(56)(7)(8) \in \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & & \\ \hline \square & & & \\ \hline \square & & & \\ \hline \end{array}$$

We saw that there exists an *one-to-one correspondence* between **irreps** and **conjugacy classes**

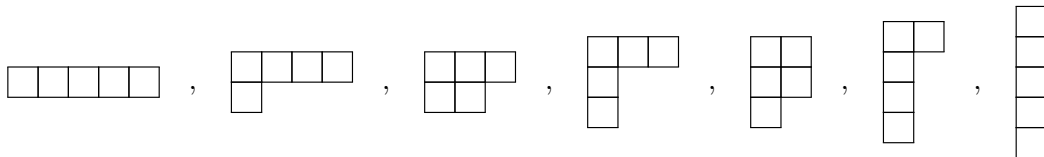
$$(\text{A tableau}) = (\text{An irrep})$$

Young Tableaux - Examples

- S_4



- S_5



Young Tableaux and Representations of S_n^*

There are $n!$ ways of putting integers from 1 to n in the boxes of the tableau

$$\begin{array}{|c|c|c|} \hline 6 & 5 & 4 \\ \hline 3 & 2 & \\ \hline 1 & & \\ \hline \end{array}, \quad \begin{array}{|c|c|c|} \hline 1 & 3 & 2 \\ \hline 4 & 5 & \\ \hline 6 & & \\ \hline \end{array}, \quad \dots$$

Since there are $n!$ states in the regular rep of S_n , we assign each tableau to a state as

$$\begin{array}{|c|c|c|} \hline 6 & 5 & 4 \\ \hline 3 & 2 & \\ \hline 1 & & \\ \hline \end{array} \text{ corresponds to } |654321\rangle$$

where $|654321\rangle$ is $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 4 & 3 & 2 & 1 \end{pmatrix} \in S_n$

Young Tableaux and Representations of S_n^*

Now we add 2 rules :

Rule 1. Symmetrize the corresponding state in the numbers in each **row**.

Rule 2. Antisymmetrize the corresponding state in the numbers in each **column**.

e.g.

$$\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array} \leftrightarrow \underbrace{(|123\rangle + |213\rangle) - (|321\rangle + |231\rangle)}_{\text{Antisymmetrized } 1 \leftrightarrow 3}$$

symmetrized $1 \leftrightarrow 2$

Young Tableaux and Representations of S_n^*

$$\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array} = |123\rangle + |213\rangle - |321\rangle - |231\rangle$$

$$\begin{array}{|c|c|} \hline 3 & 1 \\ \hline 2 & \\ \hline \end{array} = |312\rangle + |132\rangle - |213\rangle - |123\rangle = \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array} - \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}$$

$$\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array} = |132\rangle + |312\rangle - |231\rangle - |321\rangle$$

$$\begin{array}{|c|c|} \hline 3 & 2 \\ \hline 1 & \\ \hline \end{array} = |321\rangle + |231\rangle - |123\rangle - |213\rangle = -\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}$$

$$\begin{array}{|c|c|} \hline 2 & 1 \\ \hline 3 & \\ \hline \end{array} = |213\rangle + |123\rangle - |312\rangle - |132\rangle = \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array} - \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}$$

$$\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & \\ \hline \end{array} = |231\rangle + |321\rangle - |132\rangle - |312\rangle = -\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}$$

$\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}, \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}$ span all other states!

Young Tableaux and Representations of S_n^*

There are two important implications of this example

- ① Those states, constructed from , **span** a subspace of dimension 2
- ② The subspace spanned is an **invariant subspace**

Therefore for S_3 ,

- ①  : completely symmetrized states
- ②  : mixed states (multiplicity 2)
- ③  : completely antisymmetrized states

Therefore, we have reduced $D_{R,S_3} = \mathbf{1} \oplus \mathbf{2} \oplus \mathbf{2} \oplus \mathbf{1} !$

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Tensor Products

Theorem

Let V, W be finite-dimensional vector spaces. There exists a finite dimensional vector space $V \otimes W$, and a *bilinear* map $T : V \times W \rightarrow V \otimes W$ denoted

$$(v, w) \rightarrow v \otimes w, \quad (10)$$

satisfying the following properties.

- 1 If U is a vector space and $g : V \times W \rightarrow U$ is a bilinear map, then there exists a **unique linear** map

$$g_* : V \otimes W \rightarrow U$$

such that

$$g(v, w) = g_*(v \otimes w).$$

Tensor Products

Theorem

(continued)

- ② If $\{v_1, \dots, v_m\}$ is a basis of V and $\{w_1, \dots, w_n\}$ is a basis of W then the elements

$$v_i \otimes w_j$$

form a basis of $V \otimes W$.

The space $V \otimes W$ is called the **tensor product** of V and W . Note that by (2),

$$\dim V \otimes W = (\dim V)(\dim W).$$

Tensor Products

The first property states that once g is fixed, then there exists a map g_* making the following diagram commutes :

$$\begin{array}{ccc} V \times W & \xrightarrow{T} & V \otimes W \\ & \searrow g & \downarrow g_* \\ & & U \end{array}$$

This property characterizes $V \otimes W$ uniquely up to isomorphism

→ Given V and W , $V \otimes W$ is essentially **unique**

References

- Georgi, Lie Algebras in Particle Physics
- Hassani, Mathematical Physics
- Lang, Linear Algebra
- Dummit and Foote, Abstract Algebra